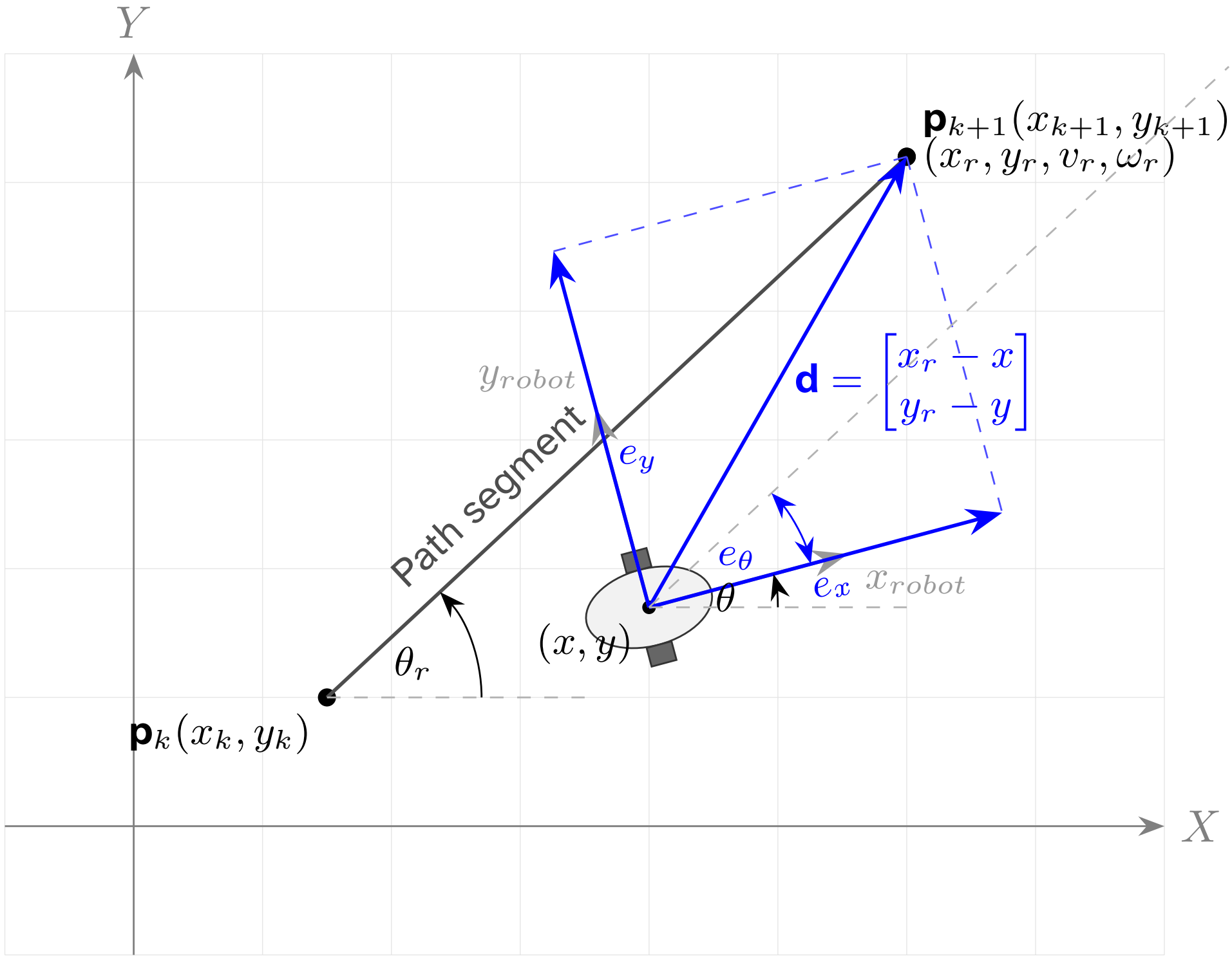

Y
X
$\mathbf{p}_{k+1}(x_{k+1}, y_{k+1})$
$(x_r, y_r, v_r, \omega_r)$
$\mathbf{d} = \begin{bmatrix} x_r - x \\ y_r - y \end{bmatrix}$
y_{robot}
e_y
Path segment
e_θ
e_x
x_{robot}
θ
(x, y)
θ_r
$\mathbf{p}_k(x_k, y_k)$